Prepare Spatial Predictors for Model Fitting

Description

Expands dataset-level spatial predictor data to the sample level by joining against the canonical '(dataset_name, age)' sample index, producing a data frame whose rows align with the community matrix and abiotic data used for model fitting.

Usage

```
prepare_spatial_predictors_for_fit(data_spatial = NULL, data_sample_ids = NULL)
```

Arguments

`data_spatial`	A data frame with 'dataset_name' as row names and one or more spatial predictor columns (e.g. 'coord_x_km', 'coord_y_km' from 'project_coords_to_metric()').
`data_sample_ids`	A data frame of valid '(dataset_name, age)' pairs as returned by 'align_sample_ids()'.

Details

Spatial predictors are stored at the dataset level (one row per site) but models require one row per sample (site × time-slice). This function replicates each dataset's spatial values across all its valid sample ages. The row-name format '"<dataset_name>__<age>"' matches that of the community matrix and abiotic data frame. Unlike 'prepare_coords_for_fit', this function is generic and imposes no assumptions on which spatial predictor columns are present.

Value

A data frame with row names in the format '"<dataset_name>__<age>"' and the same predictor columns as 'data_spatial'. Rows are sorted by 'dataset_name' then 'age', matching the ordering of the community matrix and abiotic data produced by the respective preparation functions. Rows with any 'NA' in the spatial predictors are dropped.

See Also

[project_coords_to_metric()], [align_sample_ids()], [build_jsdm_fit_input()], [prepare_coords_for_fit()]